# HUJI Applicative Research Report — Exact & Social Sciences — Week 44, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 44, 2026 · Generated October 26, 2026

**■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.**

Hebrew University researchers are advancing materials science with innovations that promise enhanced performance across multiple sectors. One significant development features carbodicarbene monolayers capable of precisely tuning the electronic properties of metal surfaces, opening doors for high-performance sensors, catalysts, and next-generation electronic components. Concurrently, a study into the robust structure of sturgeon fin bone offers bio-inspired blueprints for designing exceptionally durable and lightweight composite materials for aerospace and medical applications. These breakthroughs present compelling investment opportunities in advanced materials technology.

**RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE**

- **Elad Gross** — Materials, Clean Energy, Software/AI
- **Joshua Milgram** — Materials

## EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

## 1. Carbodicarbene Monolayers on Au(111) Enable Strong Carbon(0)-Metal Binding and Work-Function Modulation. 23/50

*"Novel carbon-metal bonding precisely tunes electronic interfaces for advanced sensors, catalysts, and next-gen electronics."*

**Main researcher: Elad Gross**

Institute of Chemistry and the Center for Nanoscience and Nanotechnology, The Hebrew University, Jerusalem, Israel.

Published in Small (Weinheim an der Bergstrasse, Germany) · 2026 Oct 23

ABOUT THIS RESEARCH

Researchers at Hebrew University have demonstrated that carbodicarbene monolayers can be assembled on Au(111) surfaces, forming exceptionally strong carbon(0)-metal bonds that allow precise tuning of the metal's work function. This surface chemistry platform offers a new route to engineer electronic properties of metal interfaces for advanced device applications.

COMMERCIAL ANGLE

The ability to modulate gold surface work function via carbodicarbene monolayers could be commercialized in high■performance sensors, catalytic converters, or next■generation electronic components where tailored surface energetics are critical.

COMMERCIALISATION METRICS

**Novelty** 8/10

The concept of carbodicarbene monolayers forming exceptionally strong carbon(0)-metal bonds and enabling tunable work-function on Au(111) represents a significant advance over existing surface chemistry, offering a potentially platform-enabling interface for electronic and catalytic applications.

**Commercial Potential** 3/10

While the fundamental surface science is elegant, it lacks a defined application path, as carbodicarbenes are highly specialized laboratory reagents with no clear advantage over existing industrial surface-functionalization technologies like SAMs. The technical complexity and stability requirements for these monolayers present significant barriers to scaling for any practical electronic or catalytic device.

**Market Size** 3/10

The research focuses on a highly specialized surface chemistry technique with limited immediate commercial applications, suggesting a modest and niche market size.

**Tech Readiness** 2/10

This is fundamental surface science research focused on the synthesis and characterization of carbodicarbene-gold interfaces, representing a very early-stage lab investigation with no defined industrial or commercial application at this time.

**IP Strength** 7/10

The study presents a novel surface chemistry that could be patented as a new class of carbodicarbene monolayers for metal work■function tuning, offering a defensible platform, but the lack of demonstrated commercial application or extensive prior art analysis limits it from a top■tier IP score.

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## 2. Functional Adaptations for Load-Bearing in a Dermal Bone: The Pectoral Fin Spine of the Russian Sturgeon (Huso Gueldenstaedtii). 10/50

*"Bio-inspired material design from sturgeon bone offers blueprints for ultra-durable, lightweight composites."*

**Main researcher: Joshua Milgram**
Robert H. Smith Faculty of Agriculture, Food and Environment, Koret School of Veterinary Medicine, Hebrew University of Jerusalem, Rehovot, Israel.
Published in Journal of morphology · 2026-10-22

ABOUT THIS RESEARCH

This research investigates the complex hierarchical microstructure and mineralization patterns of the pectoral fin spine in the Russian sturgeon to understand how it handles significant mechanical stress. By mapping the mechanical adaptations of this biological structure, the study provides a blueprint for how nature optimizes bone-like tissues for durability and load-bearing performance.

COMMERCIAL ANGLE

The structural insights from this dermal bone can be leveraged to bio-inspire the design of high-performance, lightweight, and impact-resistant composite materials for aerospace or medical implants.

COMMERCIALISATION METRICS

**Novelty** 3/10
This study focuses on descriptive morphological and biomechanical characterization of a specific biological structure, which provides incremental anatomical knowledge rather than a novel, disruptive, or platform-enabling technology. While academically valuable for biomimetics, it lacks the technical innovation or industrial scalability required for a higher novelty score in a deep-tech investment context.

**Commercial Potential** 3/10
This is a fundamental biological study of comparative anatomy and biomechanics that lacks immediate application in product development or clinical utility. While it provides bio-inspired insights, it is descriptive research rather than a translation-ready technological platform.

**Market Size** 2/10
This is a fundamental biological study of sturgeon anatomy with no clear pathway to industrial application, commercial scalability, or a defined addressable market. Its value is limited to niche academic research in evolutionary biomechanics rather than addressing a significant commercial pain point.

**Tech Readiness** 1/10
This is a fundamental biology study focusing on the structural analysis of an extant sturgeon species, providing zero clinical or industrial application potential. It offers no technological development or engineering intervention, placing it strictly in the realm of basic scientific discovery.

**IP Strength** 1/10
The paper is a descriptive biological study of anatomical adaptations in a fish species; it describes natural phenomena which are not patentable and lacks any novel engineered application or proprietary methodology.

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